

HW08

Sam Ly

March 27, 2026

Chapter 15

1. `print "${drink} will be ready in ${steep + cool} minutes.";`

What token types would you define to implement a scanner for string interpolation? What sequence of tokens would you emit for the above string literal?

Rather than defining a `TOKEN_STRING`, I would define a `TOKEN_QUOTE` and a `TOKEN_STRING_VALUE`. This way, we essentially defer some of the complex work to the parser. Then, we also need a `TOKEN_STRING_INTERP_START`

For the tokens emitted, it would be

```
TOKEN_QUOTE
TOKEN_STRING_INTERP_START
TOKEN_IDENTIFIER
TOKEN_RIGHT_BRACE
TOKEN_STRING_VALUE
TOKEN_STRING_INTERP_START
TOKEN_IDENTIFIER
TOKEN_ADD
TOKEN_IDENTIFIER
TOKEN_RIGHT_BRACE
TOKEN_STRING_VALUE
TOKEN_QUOTE
```

2. Later versions of C++ are smarter and can handle the above code. Java and C# never had the problem. How do those languages specify and implement this?

The two approaches to handling this are:

- (a) Context aware scanning/lexing.
- (b) Parser hack.

Later versions of C++ and C# use a context aware scanner, so it “knows” when it is parsing a type, and will emit the correct token. On the other hand, Java defers this work to the parser. When the parser is currently parsing a generic, it can dynamically “split” the shift operator into multiple generic closing tags.

3. Name a few contextual keywords from other languages, and the context where they are meaningful. What are the pros and cons of having contextual keywords? How would you implement them in your language’s front end if you needed to?

In modern Python, ‘match’ and ‘case’ are only keywords if they are used within a match statement. Otherwise, they can be used as variable names. Contextual keywords are useful, but in my opinion

slightly confusing. The reason Python made these keywords contextual is because of backwards compatibility. If these keywords were in from the beginning, I believe they would not be contextual. If I were to implement them, I would choose to let them be a standard identifier, then have the parser check if that identifier should be a keyword.